

CROSSVALIDATION FOR IDENTIFICATION

(I 1111)

Let consider a particular structure for the model (e.g. ARX) $M(\hat{\underline{\theta}})$
if we want to put in evid. the "complexity" $\Rightarrow$ we want to use
indices to do this.

EX: linear ARX: complexity $\equiv$ "m, m" $\Rightarrow \hat{\underline{\theta}} \rightarrow \hat{\underline{\theta}}_{m,m}$

$$\hat{y}(k|\underline{\theta}) = [1 - \hat{A}(z)] y(k) + \hat{B}(z) u(k)$$
$$\hat{\underline{\theta}} = \underbrace{[a_1 \dots a_m]}_m, \underbrace{[b_0 \dots b_m]}_{m+1}$$

O.E. complexity $\equiv$ "m, m"

ARMAX: $\equiv$ "m, m, q"

$$\hat{\underline{\theta}} \rightarrow \hat{\underline{\theta}}_{m,m,q}$$

Usually a single index is used for complexity corresponding to
 $p = \dim(\hat{\underline{\theta}})$

$$\text{ARX}_{m,m} \Rightarrow \hat{\underline{\theta}}_p \quad p = m + m + 1$$

$$\text{OE: } p = m + m + 1$$

$$\text{ARMAX: } p = m + m + q + 1$$

Obviously, using only p is not "so precise" as the "true"

complexity.

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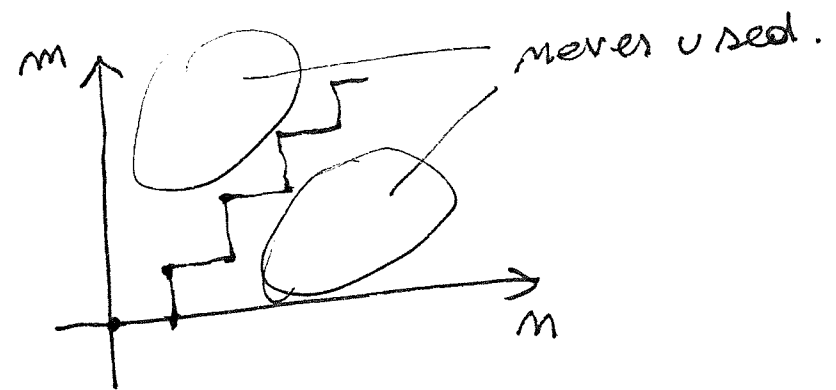
EX. ARX : $m=2, m=3 \Rightarrow p=6$

$m=1, m=4 \Rightarrow p=6$

usually some "rule" is used to denote how p increases.

EX: ARX

p	n	m
1	0	0
2	1	0
3	1	1
4	2	1
5	2	2



For simplicity let denote the complexity by $p = \dim \mathcal{G}$

We want to choose the "BEST COMPLEXITY"

~~For any complexity~~ First step.: divide the DATA SET $\begin{matrix} \swarrow \text{TS} \\ \text{VS} \\ \searrow \text{(TestSet)} \end{matrix}$

~~Usually~~ The three sets should all be "sufficiently big" because they must be "statistically representative"

Typically If only TS & VS are used
70% - 30%

(I 146)

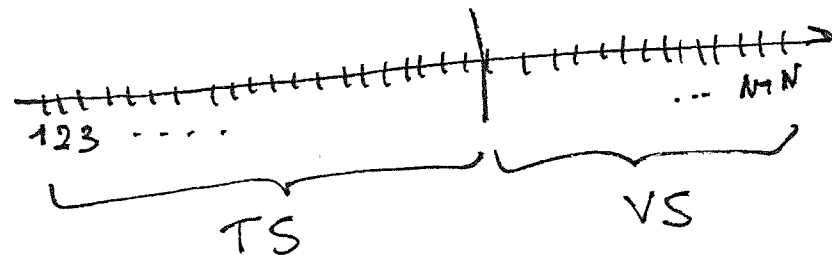
For every complexity $p = 1, 2, \dots, p_{\max}$

$$(\bullet) \quad \hat{\theta}_{-p}^* = \arg \min_{\hat{\theta}_{-p}} J_{N_{TS}}^{TS}(\hat{\theta}_{-p})$$

$$J_{N_{TS}}^{TS}(\hat{\theta}_{-p}) = \frac{1}{N_{TS}} \sum_{k \in TS} [y(k) - \hat{y}(k|\hat{\theta}_{-p})]^2$$

↓
cardin. of TS

If a single experiment is available of length N

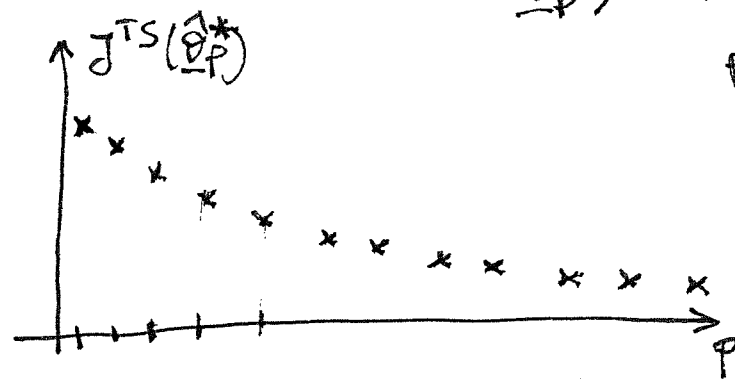


(*) corresponds to the minimization of the cost $J_{N_{TS}}^{TS}(\hat{\theta}_{-p})$

EX. ARX $\Rightarrow \hat{\theta}_{-p}^* = (\Phi_{N_{TS}} \Phi_{N_{TS}}^T)^{-1} \Phi_{N_{TS}}^T y_{N_{TS}}$

OE/ARMAX $\hat{\theta}_{-p}^*$ is obtained as the result of a
"full" application of some math. progr.
alg.

If we considered the costs $J^{TS}(\hat{\theta}_{\underline{p}}^*)$ [using the "optimized" on TS-parameters] (I 147)



BUT if p is "too big" $\Rightarrow$ OVERFITTING.

~~As~~. In any case $p = \dim \hat{\theta}$ should be smaller than N_{TS}
 $p_{max} < N_{TS}$

To choose the "best" p : p^*

$\Rightarrow$ We compute $J^{VS}(\hat{\theta}_{\underline{p}}^*)$ for all the complexities.

$$\Rightarrow p^* = \arg \min_p J^{VS}(\hat{\theta}_{\underline{p}}^*)$$

i.e. the "best" complexity is that with minimum cost on the validation set computed using the parameters optimized with TS.

"Test Set" can be used to quantify the expected cost of a model. (148)

$$\Rightarrow \underbrace{E[y(k) - \hat{y}(k|\hat{\theta}_{\underline{p}}^*)]^2}_{e^2(k|\hat{\theta}_{\underline{p}}^*)} \cong \frac{1}{N_{\text{TestSet}}} \sum_{k \in \text{TestSet}} [y(k) - \hat{y}(k|\hat{\theta}_{\underline{p}}^*)]^2$$

$$= J_{\text{TestSet}}(\hat{\theta}_{\underline{p}}^*)$$

Test Set is not necessary.

Note that the 3 sets have been chosen distinct.

because for validating we cannot use data already used for training the same for Test.

p = the complexity usually referred to as "meta-parameter".

(m, m)
 (m, m, q)

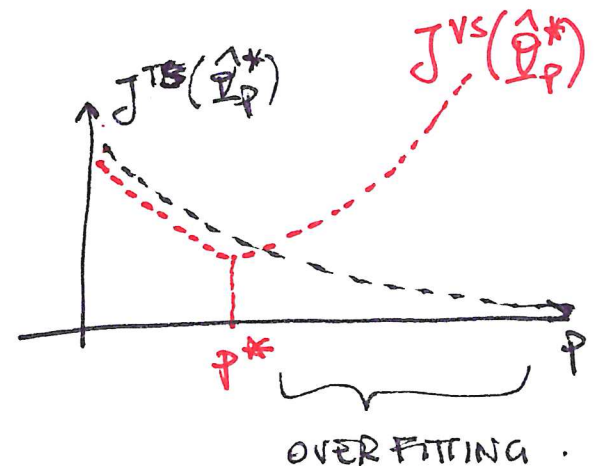
Summing up:

for $p = 1 \dots p_{\max} < N_{\text{TS}}$

$$\hat{\theta}_{\underline{p}}^* = \arg \min_{\hat{\theta}_{\underline{p}}} J^{\text{TS}}(\hat{\theta}_{\underline{p}})$$

$$J_p^{\text{VS}} = J^{\text{VS}}[\hat{\theta}_{\underline{p}}^*]$$

$$p^* = \arg \min_p J_p^{\text{VS}}$$



in a single eq:

$$\hat{p}^* = \arg \min_p J^{VS} \left[\arg \min_{\hat{\theta}_p} J^{TS}(\hat{\theta}_p) \right]$$

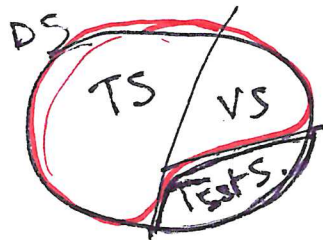
(Ilu9)

then we can compute

$$J^{\text{TestSet}}(\hat{\theta}_{\hat{p}^*}^*)$$

$\hat{\theta}_{\hat{p}^*}^*$ is param. vector where both p & $\hat{\theta}_p$ are optimized.
using VS & TS separately.

After having obtained $\hat{p}^*$: why not to optimize again $\hat{\theta}_{\hat{p}^*}$
using TS & VS together.



As the complexity is "frozen" to $\hat{p}^*$

$$\hat{\theta}_{\hat{p}^*}^{**} = \arg \min_{\hat{\theta}_{\hat{p}^*}} J^{TS+VS}(\hat{\theta}_{\hat{p}^*})$$

Test Set must remain separate if we want to
have a performance measure on the model.